Context-Aware Security and Post Quantum Cryptography applied to IoT networks

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***Abstract*—Internet of Things (IoT) devices are becoming increasingly ubiquitous in many sectors, resulting in a significant increase in the number of IoT devices. Due to the increasing availability of high-capacity storage devices at cost-effective prices, attackers can feasibly store intercepted data now and decrypt it later using advanced quantum computing techniques. In this paper, an innovative approach is presented for the efficient implementation of Post-Quantum Cryptography (PQC) in extreme edge IoT nodes. This strategy leverages context-aware security (CAS), a strategy that allows the systems to react to threats based on information collected from it environment. The use of CAS allows the network to adjust its security measures based on real-time events detected by sensors. Furthermore, the risk posed by quantum computer attacks has been assessed, leading to the design incorporating encryption algorithms that are currently considered resistant to quantum computing attacks. This paper proposes the implementation of a prototype network that employs CAS and utilizes PQC.**

***Index Terms*—Internet of Things (IoT), Post-Quantum Cryptography (PQC), Context-Aware Security (CAS), Industrial Internet of Things (IIoT)**

I. Introduction

The increasing feasibility of quantum computers becoming an integral part of our daily lives underscores the critical need to safeguard the information stored on conventional computers. This is particularly imperative because quantum computing poses a significant threat to current cryptographic methods, which may become vulnerable once quantum technology matures.

One of the primary concerns today is the potential for attackers to penetrate an organization's network, capturing data as they traverse the network. Due to the widespread availability of high-capacity storage devices at cost-effective prices, attackers can feasibly store intercepted data now and decrypt it later using advanced quantum computing techniques. This specific threat scenario is referred to as "Store Now, Decrypt Later" (SNDL).

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In response to such impending threats, the National Institute of Standards and Technology (NIST) initiated a standardization process aimed at discovering robust asymmetric encryption and digital signature algorithms capable of withstand quantum-driven attacks [1]. The outcome of this initiative has led to the adoption of several promising algorithms, including CRYSTAL Kyber [2] for key encapsulation and exchange, along with SPHINCS [3], FALCON [4], and CRYSTAL Dilithium [5] for digital signatures.

Moreover, the proliferation of Internet of Things (IoT) devices, which are now ubiquitous in many sectors, significantly expands an organization's attack surface. These devices, particularly those classified as Medical IoT (MIoT) or Industrial IoT (IIoT), often handle sensitive or confidential information and are thus prime targets for cyber attacks. Given this vulnerability, there is an urgent need to extend robust quantum-resistant cryptographic protections to these networks.

In light of these considerations, in this paper, an innovative approach is presented for the efficient implementation of Post-Quantum Cryptography (PQC) in extreme edge IoT nodes. This strategy leverages context-aware security, allowing the network to dynamically adjust the security measures of IoT nodes based on real-time events. Such an adaptive security model not only mitigates potential quantum threats, but also ensures optimal utilization of available resources, thus maintaining uninterrupted service operations.

II. Related work

A. Context-Aware Security

Context-Aware Security (CAS) is a strategy by which the environment of a system is analyzed and various security measures are applied based on this analysis. The research documented in [7] illustrates that as vehicles evolve from simple mechanisms to complex computer systems, it is necessary to implement new security measures, including those based on the context of the vehicle.

Furthermore, in [6] the new 6G mobile telephony standard can utilize physical-level features with low latency, low energy consumption, and low complexity to enhance network security

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using CAS. This highlights a significant advance in the integration of context-aware strategies into modern communication infrastructures.

In addition, in [8] the potential of equipping vehicles with security information and event management (SIEM) systems is explored, improving the effectiveness of decision-making even in vehicles lacking Internet connectivity. The information collected is then shared with other vehicles, promoting a collaborative approach to vehicular security.

CAS is also applicable in sensor networks. As shown in [9], CAS can be implemented using Artificial Intelligence (AI) in a variety of IoT devices found in modern connected homes. The AI model's output is subsequently integrated into a Suricata rule engine that ensures network security.

In addition, in [10] is explored how network security policies can be modified based on contextual information on network usage, thus altering and ensuring the enforcement of new security policies at each device access point.

Lastly, in [11] is proposed the use of a real-time learning method with low computational overhead, suitable for implementation at nodes to prevent misuse and intrusion into sensor networks.

However, none of the referenced implementations utilize sensor measurements to switch between different encryption protocols within the network, nor do they employ Post-Quantum Cryptography (PQC) algorithms to secure the network against quantum threats. This indicates a potential area for further research and development in the field of CAS systems, particularly in improving resilience against advanced quantum computing risks.

B. Post-Quantum Cryptography on IoT devices

Post-Quantum Cryptography is a solution derived from the necessity to protect information from a possible materialization of a quantum threat, even on low-resource devices such as IoT devices.

In [12] there is an implementation of different PQC algorithms for the Cortex-M4 instruction set. In this library, every PQC algorithm submitted to the NIST standardization process is implemented for testing on devices that use the Cortex M4 instruction set.

In [13] the possibility of implementing one of the candidates PQC algorithms natively using the mobile instruction set is explored, achieving an improvement of more than x2 in specific parts of the algorithm, thus contributing to an overall improvement to the algorithm speed.

In [14] the NTRU PQC algorithm is analyzed using an extreme IoT device with very low resources, demonstrating that the PQC algorithm can be used on IoT devices with low performance impact.

In addition, in [15] the Saber finalist PQC algorithm is used to provide an improved authentication scheme for MIoT devices and compare this new scheme with the previous version based on lattice-cryptography.

III. Implementation

This paper proposes the implementation of a prototype network utilizing CAS and PQC. The network incorporates various security measures that are toggled based on environmental data from sensors embedded in IoT devices. This method supports a dynamic security protocol tailored to specific conditions and threats identified in real-time by the IoT environment, thus improving the network's security infrastructure in terms of both responsiveness and robustness.

The implementation aims to safeguard the network against quantum threats, such as quantum computers and SNDL threats, by using well-known quantum-resistant cryptographic methods. A mesh topology is deployed in this implementation, assuming a higher energy consumption for nodes closer to the IoT network root due to packets' retransmissions from all nodes relaying messages through a specific node.

A. IoT device

To implement this prototype, the device known as Cookie [16] (see Figure 1), which is considered an extreme edge device with limited capabilities, has been used. The key features of this device are highlighted below:

- SoC: EFR32MG12P432F104GL125 @ 40 MHz
- RAM: 256 kB
- ROM: 1024 kB
- Connectivity: Bluetooth and IEEE 802.15.4
- Sensors: Accelerometer, Gyroscope, Temperature, Humidity, and Energy Consumption
- OS: Contiki-NG v4.9 [17]

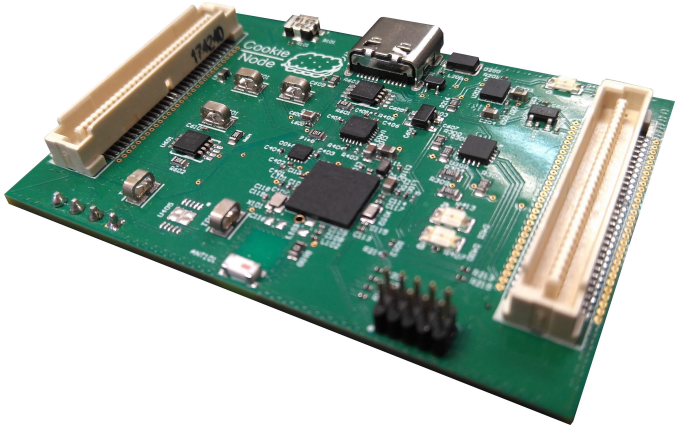


Fig. 1. Cookie sensor

The use of Contiki-NG facilitates the implementation of other network protocols such as the Routing Protocol for Low-Power and Lossy Networks (RPL), IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN), and the User Datagram Protocol (UDP). These protocols are used because of their capability to sustain a network in a hostile environment where packet loss happens a lot and because of their reduced energy consumption.

Using the RPL protocol generates a tree topology network 2, where one of the IoT nodes acts as the root or coordinator,

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and the others connect as leaf nodes, if they do not have any other nodes to relay messages to, or intermediate nodes, if they need to forward messages from other nodes in the network.

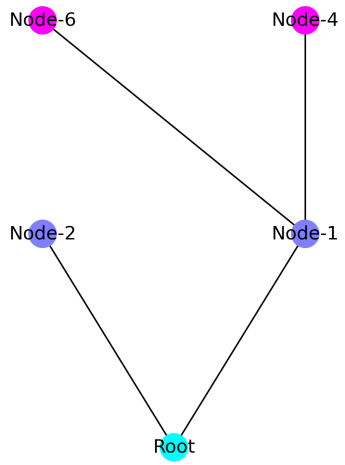


Fig. 2. Logical representation of the IoT network employed for this experiment

For testing the application, five Cookie devices were deployed, one of which acted as the network coordinator, while the others served as clients within the network. These devices were left operational for almost two hours, randomly placed on a laboratory table without external intervention until the experiment started to be repetitious.

B. Context Aware Security Prototype Implementation

For the implementation of this prototype, a series of UDP messages are exchanged between the client devices on the network and the coordinator using the protocol stack depicted in Figure 3.

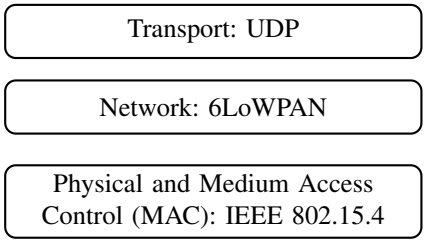


Fig. 3. Protocol stack used by the prototype

Upon establishing communication between the coordinator and the clients, the next step is to define the level of security to be employed in the communications of the prototype. Different security levels are outlined in Figure 4, where each successive level incorporates the security measures of the previous levels along with new measures:

- Level 0: at this level, only network-level security present in the IEEE 802.15.4 protocol is employed, which involves using a pre-shared 128-bit Advanced Encryption Standard (AES) key. This level is activated when the network's energy consumption exceeds 550 mW.
- Level 1: in addition to network-level security, a 256-bit AES pre-shared key is used among all network nodes to encrypt information shared between nodes. This level is activated when the network's average energy consumption is between 450 and 550 mW.
- Level 2: this level employs a 512-bit Kyber as a Key Encapsulation Mechanism (KEM) between the coordinator and each of the network clients. The secret derived is subsequently used to encrypt communications using AES-256. This level is activated when the network's average energy consumption is below 450 mW.

These levels were chosen after observing how the energy consumption of the node changes while executing their code and evaluating different values to change security levels. Other levels only allowed the prototype to work on one security level without allowing it to change.

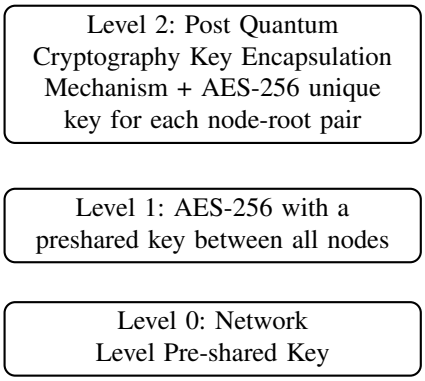


Fig. 4. Summary of prototype security levels

This prototype aims to maximize the operations of the IoT device network while maintaining low energy consumption. To this end, the network coordinator receives energy consumption measurements from each of the network nodes and calculates a weighted average among all network nodes, the coordinator and the range of each node within the network. Equation 1 shows the formula used.

$$\sum_{i=0}^{n} \frac{x_i p_i}{\sum_{i=0}^{n} p_i} \tag{1}$$

Where:

- x: is the consumption value collected by the nodes.
- p: is the weight assigned to each of the nodes.

Given that this is a weighted average and there is no clearly identified method of weight allocation, it is necessary to generate our own weights. Equation 2 shows how the specific weight of a network node is calculated. Where:

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- max_rank: is the maximum rank of the nodes in the network.
- $node_rank$: is the rank of the node for which the weight is calculated.

$$p_i = \frac{max_rank - (node_rank - 1)}{max_rank} \quad (2)$$

IV. Results

To evaluate the results obtained from the experimentation, in this paper, the duration of the IoT node network to remain operational at the highest possible security level will be evaluated. By design, the network always starts at the highest available level, which in this case involves using Post-Quantum Cryptography (PQC) for the exchange of symmetric keys and AES-256 for encrypting communications.

It is also necessary to monitor the average energy consumption of the network, since this metric will indicate whether a change in the network security level is required.

Figure 5 illustrates how the network's energy consumption evolves from the moment the root node is initiated. Please note that each time the network's consumption drops below 300 mW is due to a forced reboot of the root node, which necessitates that the rest of the nodes repeat the KEM process so can be tested how to prototype works after a software reboot.

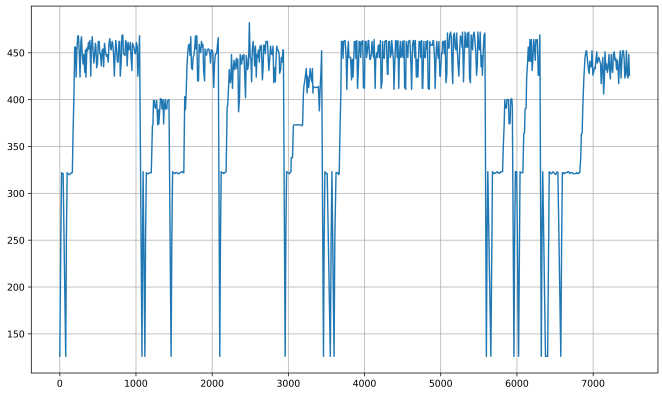


Fig. 5. Average energy consumption of the IoT network

It is evident that the average network consumption increases at the moment data exchange begins between the clients and the server, as the consumption of each node that has successfully established communication with the server is included in the network's total consumption.

As the average network consumption changes, the security level used should be adjusted accordingly. Figure 6 shows how the security level of the network varies as the average energy consumption exceeds the threshold established for that security level.

To prevent frequent changes and to avoid losing important information sent from clients to the root, the network is restricted from changing security levels for the next 100 seconds, which encourages all nodes to reduce their consumption in a uniform and stable manner.

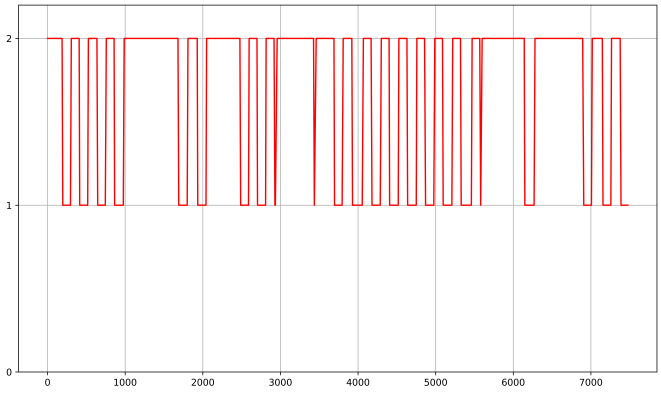


Fig. 6. Security level of the IoT network

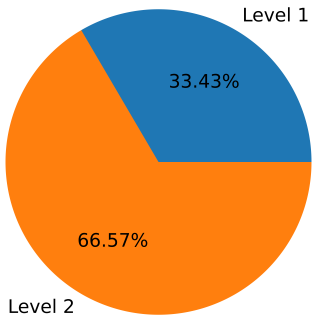


Fig. 7. Security levels pie chart

After executing the experiment, it is observed that the network remained 66. 57% of the time at the highest available security level (in this case, level 2) and only 33.43% of the time at a lower security level, allowing the network to conserve energy during periods of higher energy consumption.

The minimal security level remains inactive because the network's average energy consumption is below the required threshold to trigger it, which in this instance is security level 0.

V. Discussion

The study highlights the integration of Context-Aware Security (CAS) and Post-Quantum Cryptography (PQC) within IoT networks to mitigate threats posed by quantum computing. A prototype system was designed to dynamically modify security levels based on the energy consumption of nodes, utilizing various encryption methods. Findings indicated that the network sustained its highest security level for 66.57% of the time, showcasing effective energy management and resilience against attacks.

This approach is novel, as it leverages sensor data to transition between encryption protocols, including PQC, to counter quantum threats. However, some limitations were observed, such as potential vulnerabilities during lower security levels,

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although such attacks would require in-depth knowledge of the system, thus diminishing the likelihood of their occurrence.

VI. Conclusion

A new method has been introduced that enables IoT device networks to utilize context-aware security (CAS), allowing adjustments to network security based on environmental data collected by sensors. This method aims to maximize the resources available to the network, thus extending its useful life.

A CAS prototype has been created in this scenario, encompassing multiple security tiers, each adding more safeguards than the preceding level. Additionally, the threat of quantum-based attacks has been evaluated, influencing the inclusion of encryption algorithms believed to be resistant to quantum computing and Store Now, Decrypt Later threats. Moreover, leveraging CAS enables the system to withstand conventional privacy challenges.

To enable experimentation, a prototype with five nodes was set up. These nodes tracked their energy usage during operation, and this information was utilized to modify the network's security level, ensuring that node consumption remained controlled and resources were conserved. In the future, other node measurements will be considered to ensure that changes in the security level are more accurately aligned with the IoT network's actual real-world conditions, including external temperature.

Despite the very positive results presented, this work is not free of limitations. There is a trade-off between security and service availability. It could be possible for an attacker to target the system while it is working on a security lower level, easing the attack and compromising the system. Although this is possible, the attacker must know how the system works and when the system is operating at a lower security level, making the attack much less possible.

In the future, there are plans to improve the capabilities of the prototype and test its performance with these new features. One such future capability includes the ability to digitally sign messages exchanged by the network's sensors, using both traditional and quantum-resistant signature algorithms.

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References

[1] Alagic, G., Apon, D., Cooper, D., Dang, Q., Dang, T., Kelsey, J., Lichtinger, J., Liu, Y.-K., Miller, C., Moody, D., Peralta, R., Perlner, R., Robinson, A., & Smith-Tone, D. (2022). Status report on the third round of the NIST Post-Quantum Cryptography Standardization process. https://doi.org/10.6028/NIST.IR.8413-upd1

[2] Avanzi, R., Bos, J., Ducas, L., Kiltz, E., Lepoint, T., Lyubashevsky, V., ... & Stehlé, D. (2017). Crystals-kyber. NIST, Tech. Rep.

[3] Bernstein, D. J., Hülsing, A., Kölbl, S., Niederhagen, R., Rijneveld, J., & Schwabe, P. (2019, November). The SPHINCS+ signature framework. In Proceedings of the 2019 ACM SIGSAC conference on computer and communications security (pp. 2129-2146).

[4] Falcon (Online). Available: https://falcon-sign.info/

[5] Ducas, L., Kiltz, E., Lepoint, T., Lyubashevsky, V., Schwabe, P., Seiler, G., & Stehlé, D. (2018). Crystals-dilithium: A lattice-based digital signature scheme. IACR Transactions on Cryptographic Hardware and Embedded Systems, 238-268.

[6] Chorti, A., Barreto, A. N., Köpsell, S., Zoli, M., Chafii, M., Sehier, P., Fettweis, G., & Poor, H. V. (2022). Context-Aware Security for 6G Wireless: The Role of Physical Layer Security. IEEE Communications Standards Magazine, 6(1), 102–108. https://doi.org/10.1109/MCOMSTD.0001.2000082

[7] T. Kumari, A. Rakib, A. Zaslavsky, H. Jadidbonab & V. Moghaddam, "A Context-Aware Framework for Analysing Automotive Vehicle Security," 2024 IEEE 18th International Conference on Semantic Computing (ICSC), Laguna Hills, CA, USA, 2024, pp. 81-88, doi: 10.1109/ICSC59802.2024.00019.

[8] M. Martín-Pérez et al., "A Testbed for a Nearby-Context Aware: Threat Detection and Mitigation System for Connected Vehicles," 2023 JNIC Cybersecurity Conference (JNIC), Vigo, Spain, 2023, pp. 1-8, doi: 10.23919/JNIC58574.2023.10205891.

[9] Noor, Z., Hina, S., Hayat, F., & Shah, G. A. (2023). An intelligent context-aware threat detection and response model for smart cyber-physical systems. Internet of Things, 23, 100843.

[10] Inshi, S., Chowdhury, R., Ould-Slimane, H., & Talhi, C. (2023). Secure adaptive context-aware ABE for smart environments. IoT, 4(2), 112-130.

[11] Krinkin, K. (2023). On-device context-aware misuse detection framework for heterogeneous IoT edge. Applied Intelligence, 53(12), 14792-14818.

[12] Kannwischer, M. J., Rijneveld, J., Schwabe, P., & Stoffelen, K. (2019). pqm4: Testing and Benchmarking NIST PQC on ARM Cortex-M4.

[13] Sim, M., Kwon, H., Eum, S., Song, G., Lee, M., & Seo, H. (2023, August). Efficient Implementation of the Classic McEliece on ARMv8 Processors. In International Conference on Information Security Applications (pp. 324-337). Singapore: Springer Nature Singapore.

[14] Señor, J., Portilla, J., & Mujica, G. (2022). Analysis of the ntru post-quantum cryptographic scheme in constrained iot edge devices. IEEE Internet of Things Journal, 9(19), 18778-18790.

[15] Adeli, M., Bagheri, N., Maimani, H. R., Kumari, S., & Rodrigues, J. J. (2023). A Post-Quantum Compliant Authentication Scheme for IoT Healthcare Systems. IEEE Internet of Things Journal.

[16] Merino, P., Mujica, G., Señor, J., & Portilla, J. (2020). A modular IoT hardware platform for distributed and secured extreme edge computing. Electronics, 9(3), 538.

[17] Oikonomou, G., Duquennoy, S., Elsts, A., Eriksson, J., Tanaka, Y., & Tsiftes, N. (2022). The Contiki-NG open-source operating system for next generation IoT devices. SoftwareX, 18, 101089

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